



Laboratorio 2

1. Base de Datos:

Link imágenes:  imagenes_output

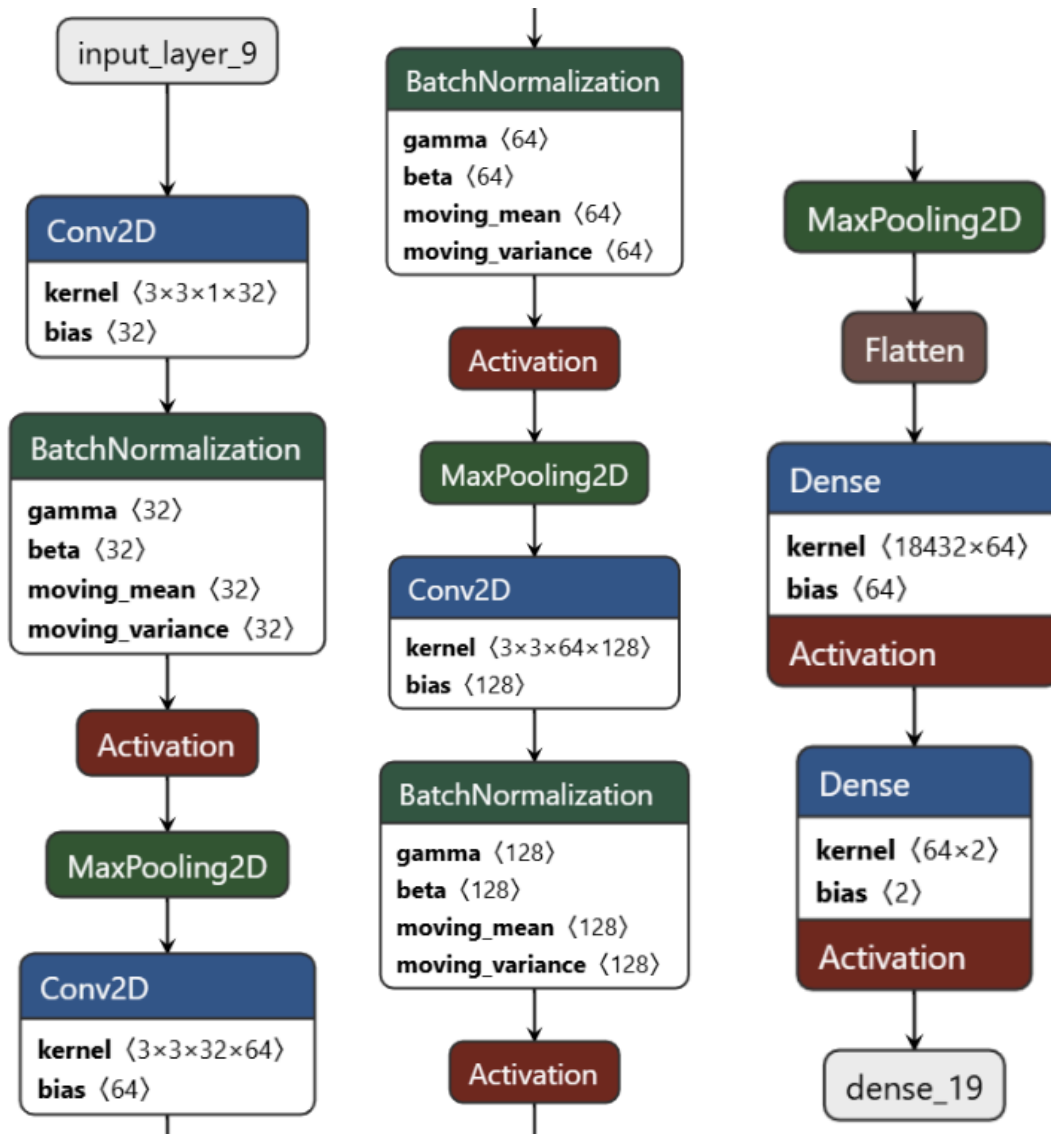
Link en plano por redundancia

<https://drive.google.com/drive/folders/12mspXe9Wtfe-LQgij4JqDxORMJKWFfFz?usp=sharing>

Contactar maferes@miuandes.cl en caso de fallos con el link.

2. Modelos

Topología 1 (CNN Keras)



Accuracy: 96.34%

Número de Operaciones:

Bloque1 = 5.308.416

Bloque2 = 84.934.656

Bloque3 = 84.934.656

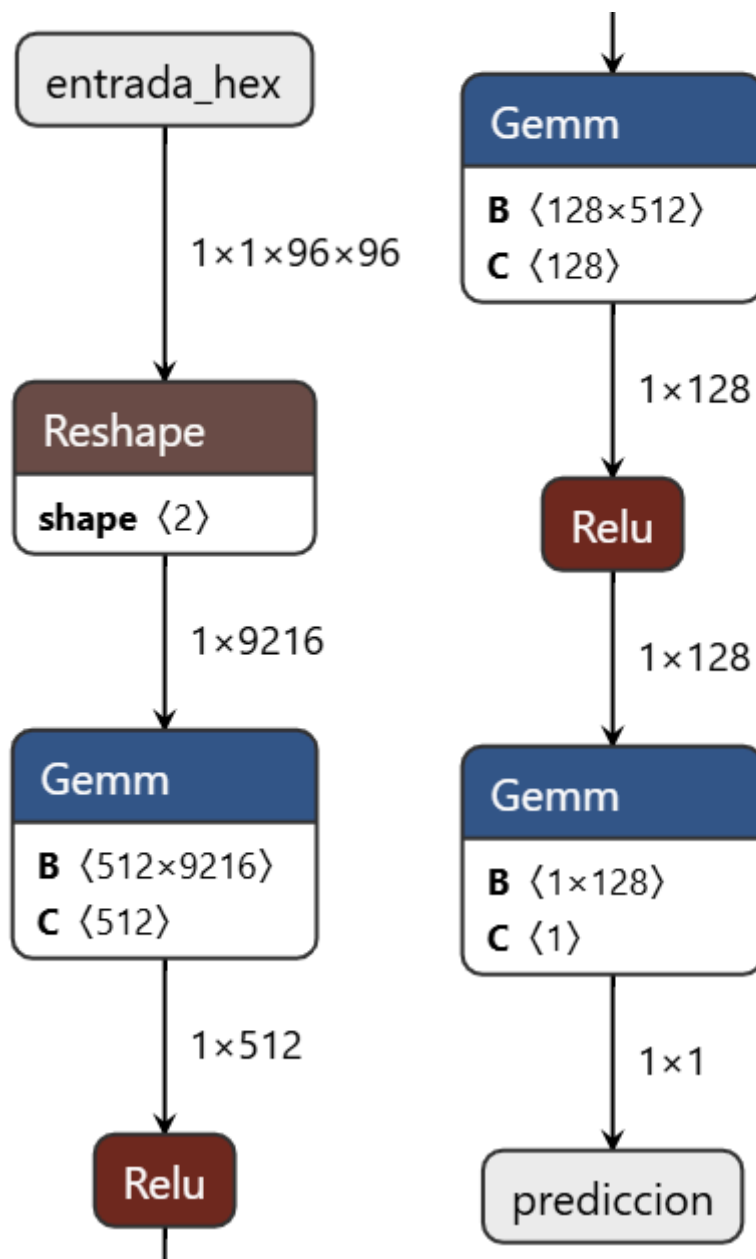
Dense.1 = 2.359.296

Dense.2 = 256

Total de operaciones = 177.537.280

Tamaño: 4.86 MB

Topología 2 (MLP Simple)



Accuracy: 77.317%

Número de operaciones:

Linear.1 = 9.437.184

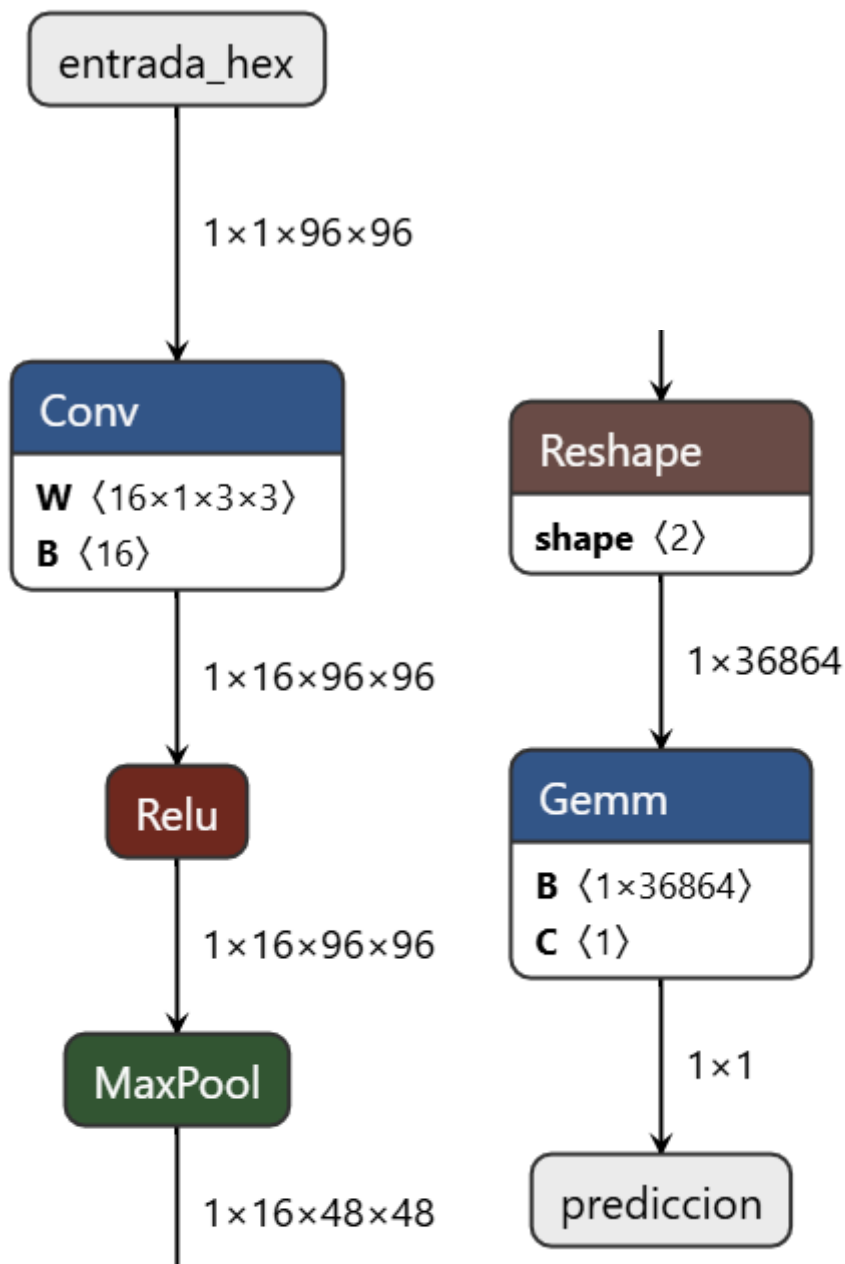
Linear.2 = 65.536

Linear.3 = 128

Total de operaciones = 9.502.848

Tamaño: 18.25 MB

Topología 3 (CNN Simple)



Accuracy: 99.51%

Número de operaciones:

Conv2D.1 = 2.654.208

ReLU = 147.456

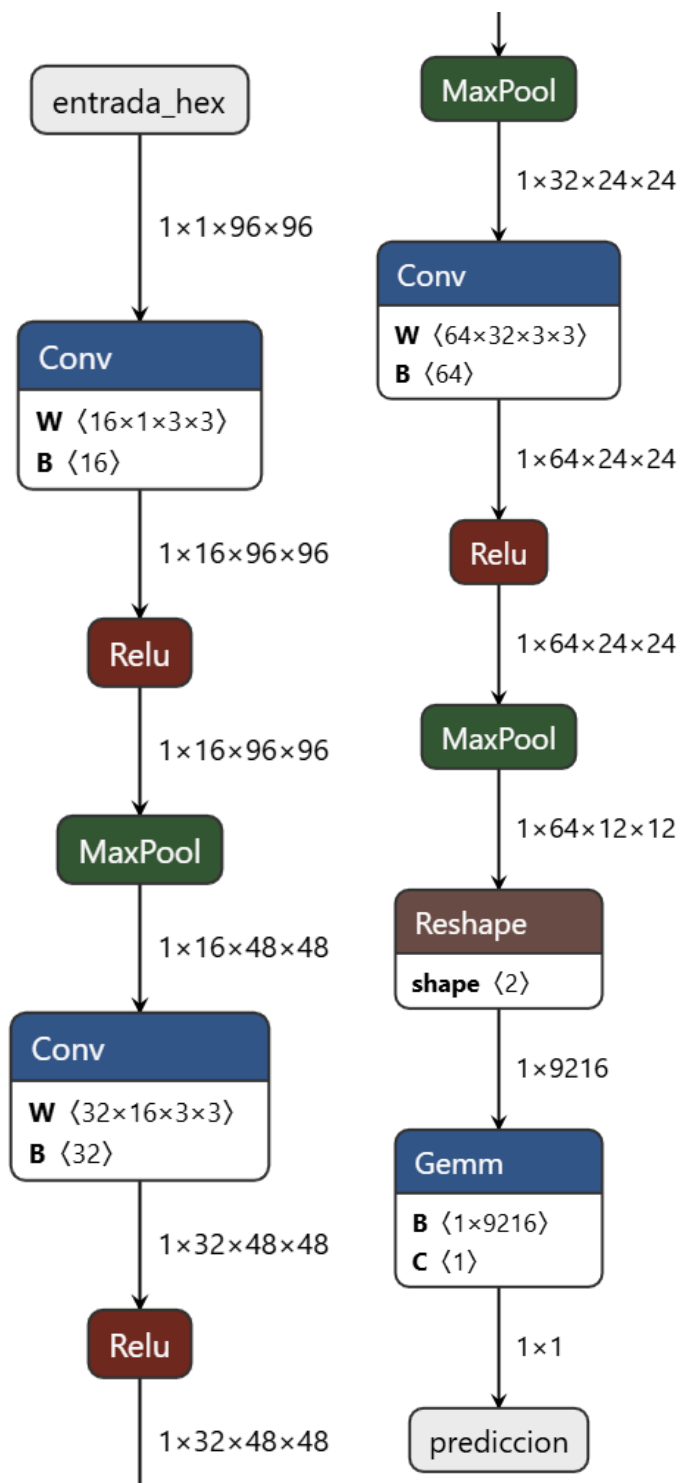
MaxPool = 147.456

Linear = 73.728

Total de operaciones: 3.022.848

Tamaño: 144.6 KB

Topología 4 (CNN Profundo)



Accuracy: 99.51%

Número de operaciones:

Bloque1 = 2.654.208

Bloque2 = 21.233.664

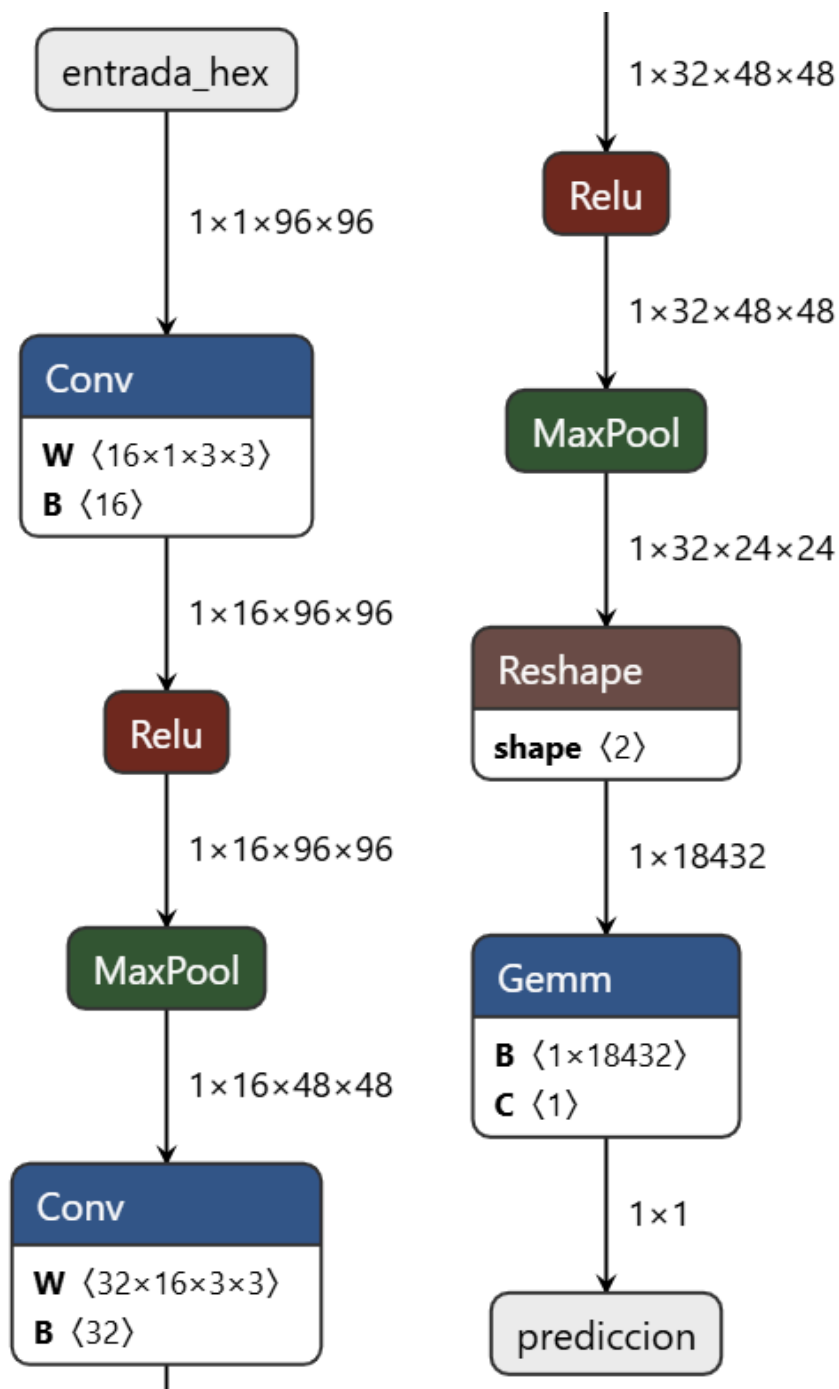
Bloque3 = 21.233.664

Linear = 18.432

Total de operaciones = 45.139.968

Tamaño: 91.5 KB

Topología 5 (CNN con Dropout)



Accuracy: 96.34%

Número de operaciones:

Bloque1 = 2.654.208

Bloque2 = 21.233.664

Linear = 36.864

Total de operaciones = 23.924.736

Tamaño: 90.7 KB

3. Estimación

output de esp32s3:

```
I (385) BENCH: =====
I (385) BENCH:   BENCHMARK ESP32-S3 - Lab 2 Sección 2.3
I (395) BENCH:   CPU: 240 MHz | SRAM: 512 KB
I (395) BENCH: =====
I (905) BENCH:
--- Calibracion hardware ---
I (905) BENCH: Tiempo por MAC   : 0.1754 us   (5.70 MMACS/s)
I (925) BENCH: Tiempo por byte : 3.971 ns    (Ancho de banda: 251.8 MB/s)
I (1125) BENCH:
--- Resultados por red ---
I (1125) BENCH: Red                               Lat (ms)    FPS    FLOPs (M)    Mem (MB)    Pot (mW)
Mod(KB)
I (1125) BENCH: -----
-----
I (1125) BENCH: MLP Simple                               1090      0.9      7.09      0.33      87.8
13857.3
I (1145) BENCH: CNN Simple                               268      3.7      1.29      0.33      72.9
577.3
I (1155) BENCH: CNN Keras                               2112     0.5     31.56      2.47     106.4
4628.6
I (1165) BENCH: CNN Profundo                               1772     0.6     15.19      1.44     100.2
1176.3
I (1175) BENCH: CNN con Dropout                               970      1.0     10.32      2.18      85.6
4610.1
I (1185) BENCH:
--- Detalle por capas ---
I (1185) BENCH:
[MLP Simple] lat=1090 ms  fps=0.9
I (1185) BENCH:   Capa                               Params      MACs      In(KB)      Out(KB)
I (1185) BENCH:   Flatten                               0              0      108.0      108.0
I (1195) BENCH:   Dense 128                     3539072     3538944      108.0       0.5
I (1205) BENCH:   Dense 64                        8256        8192        0.5        0.2
I (1205) BENCH:   Dense 2                          130          128        0.2        0.0
I (1215) BENCH:   TOTAL                     3547458     3547264      216.8     108.8
I (1225) BENCH:   Mem movida total: 0.33 MB | FLOPs: 7.09 M | Modelo: 13857.3 KB
I (1255) BENCH:
[CNN Simple] lat=268 ms  fps=3.7
I (1255) BENCH:   Capa                               Params      MACs      In(KB)      Out(KB)
I (1255) BENCH:   Conv2D 8x3x3                     224      497664      108.0       72.0
I (1255) BENCH:   MaxPool2D                          0              0       72.0      18.0
I (1265) BENCH:   Flatten                               0              0       18.0      18.0
I (1275) BENCH:   Dense 32                     147488     147456       18.0        0.1
I (1285) BENCH:   Dense 2                          66          64        0.1        0.0
I (1285) BENCH:   TOTAL                     147778      645184      216.1     108.1
I (1295) BENCH:   Mem movida total: 0.33 MB | FLOPs: 1.29 M | Modelo: 577.3 KB
I (1325) BENCH:
[CNN Keras] lat=2112 ms  fps=0.5
I (1325) BENCH:   Capa                               Params      MACs      In(KB)      Out(KB)
I (1325) BENCH:   Conv2D 16x3x3                     448     3981312      108.0     576.0
I (1325) BENCH:   MaxPool2D                          0              0      576.0     144.0
I (1335) BENCH:   Conv2D 32x3x3                     4640    10616832      144.0     288.0
I (1345) BENCH:   MaxPool2D 2                          0              0      288.0       72.0
I (1355) BENCH:   Flatten                               0              0       72.0       72.0
I (1355) BENCH:   Dense 64                     1179712     1179648       72.0        0.2
I (1365) BENCH:   Dense 2                          130          128        0.2        0.0
I (1375) BENCH:   TOTAL                     1184930    15777920     1260.2    1152.3
I (1375) BENCH:   Mem movida total: 2.47 MB | FLOPs: 31.56 M | Modelo: 4628.6 KB
I (1405) BENCH:
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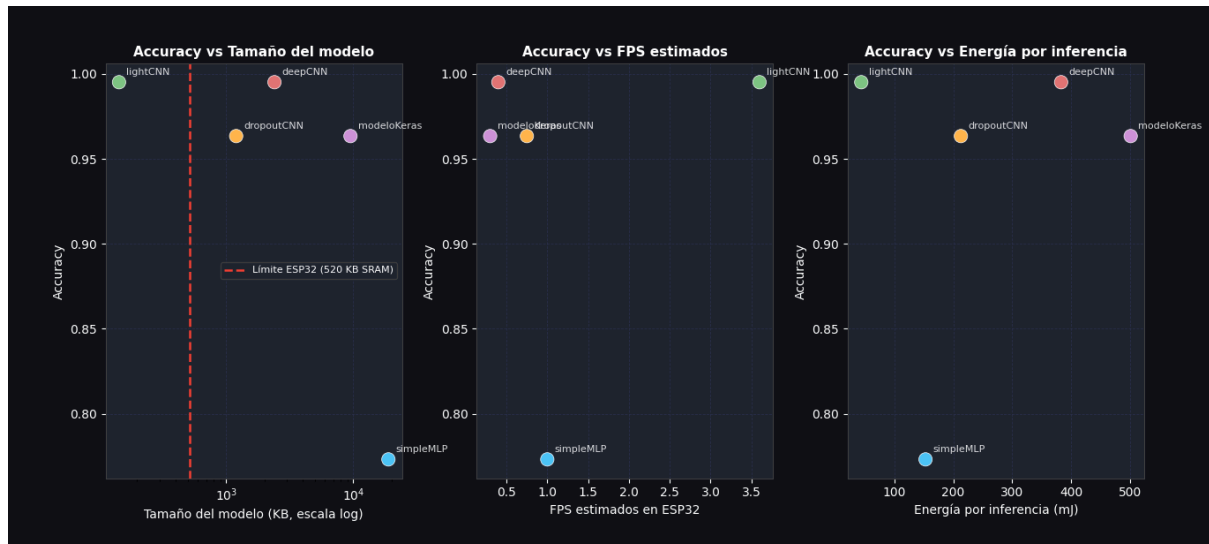
[CNN Profundo] lat=1772 ms fps=0.6
I (1405) BENCH:  Capa                Params      MACs      In(KB)    Out(KB)
I (1405) BENCH:  Conv2D 8x3x3          224          1990656    108.0     288.0
I (1405) BENCH:  MaxPool2D             0              0     288.0     72.0
I (1415) BENCH:  Conv2D 16x3x3        1168          2654208    72.0     144.0
I (1425) BENCH:  MaxPool2D 2           0              0     144.0     36.0
I (1435) BENCH:  Conv2D 32x3x3        4640          2654208    36.0     72.0
I (1435) BENCH:  MaxPool2D 3           0              0     72.0     18.0
I (1445) BENCH:  Flatten                0              0     18.0     18.0
I (1455) BENCH:  Dense 64              294976          294912    18.0      0.2
I (1455) BENCH:  Dense 2                130             128      0.2      0.0
I (1465) BENCH:  TOTAL                 301138          7594112   756.2    648.3
I (1475) BENCH:  Mem movida total: 1.44 MB | FLOPs: 15.19 M | Modelo: 1176.3 KB
I (1505) BENCH:
[CNN con Dropout] lat=970 ms fps=1.0
I (1505) BENCH:  Capa                Params      MACs      In(KB)    Out(KB)
I (1505) BENCH:  Conv2D 16x3x3         448          3981312    108.0     576.0
I (1505) BENCH:  MaxPool2D             0              0     576.0    144.0
I (1515) BENCH:  Dropout 0.3           0              0     144.0    144.0
I (1525) BENCH:  Flatten                0              0     144.0    144.0
I (1535) BENCH:  Dense 32              1179680          1179648    144.0      0.1
I (1535) BENCH:  Dropout 0.5           0              0       0.1      0.1
I (1545) BENCH:  Dense 2                 66             64       0.1      0.0
I (1555) BENCH:  TOTAL                 1180194          5161024   1116.2   1008.3
I (1555) BENCH:  Mem movida total: 2.18 MB | FLOPs: 10.32 M | Modelo: 4610.1 KB
I (1585) BENCH:
=====
I (1585) BENCH:  BENCHMARK COMPLETADO
I (1585) BENCH:  =====

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Tabla de Estimaciones			
	Latencia	Potencia	FPS estimado
CNN Keras	2112 (ms)	106.4 (mW)	0.5
MLP Simple	1,090 (ms)	87.8 (mW)	0.9
CNN Simple	268 (ms)	72.9 (mW)	3.7
CNN Profundo	1772 (ms)	100.2 (mW)	0.6
CNN con Dropout	970 (ms)	85.6 (mW)	1.0

4. Análisis

Gráficos Análisis de redes



Debido a su tamaño, la única red que puede funcionar sin memorias externas, es light CNN, esto a pesar de que sospechamos que su accuracy se puede deber a un caso de overfitting. Esta red de todas formas muestra un buen desempeño en todas las métricas usadas para medirla por lo que es definitivamente la mejor opción.